

Generative AI in education and research: A systematic mapping review

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Abstract

Given the potential applications of generative AI (GenAI) in education and its rising interest in research, this systematic review mapped the thematic landscape of 407 publications indexed in the Web of Science, ScienceDirect and Scopus. Using EPPI Reviewer, publication type, educational level, disciplines, research areas and applications of GenAI were extracted. Eight discursive themes were identified, predominantly focused on 'application, impact and potential', 'ethical implication and risks', 'perspectives and experiences', 'institutional and individual adoption', and 'performance and intelligence'. GenAI was conceptualised as a tool for 'pedagogical enhancement', 'specialised training and practices', 'writing assistance and productivity', 'professional skills and development', and as an 'interdisciplinary learning tool'. Key gaps highlighted include a paucity of research and discussions on GenAI in K-12 education; a limited exploration of GenAI's impact using experimental procedures; and a limited exploration of the potential and ethical concerns of GenAI from the lens of cultural dimensions. Promising opportunities for future research are highlighted.

KEYWORDS

ChatGPT, GenAI, higher education, large language models, school

Context and implications

Rationale for this study

The emergence of GenAI has attracted substantial attention from researchers, educational institutions, policy makers and the wider public. However, due to stringent review protocols, existing review articles examine a limited number of articles. We address this gap by employing broader inclusion criteria, leading to comprehensive findings.

Why the new findings matter

The analysis identified eight primary research domains extensively explored within the article corpus, alongside five thematic constructs elucidating the conceptualisation of GenAI applications. These insights hold significance in guiding the advancement and adoption of GenAI technologies to augment pedagogy, learning and research funding policies.

Implications for researchers and educational institutions

Due to a growing research interest since the emergence of GenAI, there are opportunities for further review studies to be conducted to uncover more future trends. First, there are research opportunities in K-12 education to understand the impact of GenAI at this level. Additional research opportunities include the exploration of cultural influences to understand GenAI adoption, and the investigation of impacts to inform policy and funding. As a majority of the extracted articles indicated ethical concerns, educational institutions should enforce ethical guidelines to encourage responsible use of GenAI.

INTRODUCTION

In recent years, the emergence of artificial intelligence (AI) technologies has significantly transformed education and research. Among these advancements, generative AI (GenAI) technologies stand out as a promising frontier that offers novel capabilities to autonomously generate new content, such as text, images and even multimedia (Tlili et al., 2023). The potential applications of GenAI technologies in education are multifaceted (Yusuf et al., 2024). For instance, intelligent tutoring systems powered by generative models can dynamically generate personalised learning materials tailored to individual student needs (Humphry & Fuller, 2023; Leng, 2024). Furthermore, they can aid in developing virtual laboratories and simulations (Araújo & Saúde, 2024), enabling students to engage in hands-on learning experiences without physical constraints. In research, generative models have shown potential utility in academic writing, offering opportunities for automated content generation, citation suggestion and language enhancement (Lim et al., 2023).

The widespread proliferation of GenAI has attracted considerable attention in scholarly publications, with various attempts made to review the patterns and trends of publications within the educational context. However, aside from the review by Bahroun et al. (2023) that included a corpus of 217 articles, most reviews have rarely included a large number of studies. On average, only 41.1 studies were included across 10 different evidence syntheses

(Ansari et al., 2023; Ipek et al., 2023; Kim et al., 2023; Labadze et al., 2023; Liu et al., 2023; Memarian & Doleck, 2023; Montenegro-Rueda et al., 2023; Pradana et al., 2023; Vargas-Murillo et al., 2023; Zirar, 2023). The smaller number of articles considered in these reviews is largely due to stringent inclusion criteria. However, this narrow selection raises critical questions about the comprehensiveness and representativeness of the current discourse surrounding GenAI's impact on education.

While these reviews acknowledge this constraint, they collectively advocate for future studies to incorporate more publications to enhance the depth of existing literature. As the academic discourse surrounding GenAI continues to rapidly evolve, it becomes evident that there is a pressing need to gain an overarching view of research undertaken in the field to date, in order to understand where further research could provide the most insight. Recognising this need, our study aims to contribute to the ongoing dialogue by mapping key themes and trends, thereby enriching the scholarly understanding of the multifaceted implications and trajectories associated with GenAI in education, and providing valuable insights for policy makers, educators and researchers, informing the development and implementation of GenAI technologies to enhance teaching and learning outcomes.

LITERATURE REVIEW

Conceptual understanding of GenAI

GenAI incorporates a diverse range of artificial intelligence models that generate unique, human-like content across various media formats, such as text, images, audio and video (Yusuf et al., 2024). Unlike traditional pattern recognition or rule-based methods, GenAI models are designed to mimic the creative and generative aspects of human cognition (Dwivedi et al., 2023). These systems mainly utilise machine learning methodologies, including deep learning and neural networks, to learn from extensive datasets (Tlili et al., 2023). Through rigorous training, they acquire the ability to identify underlying patterns and structures within the data. Additionally, they employ probabilistic models to produce new content that aligns with these learned patterns, resulting in coherent and contextually relevant outputs (Tlili et al., 2023).

Previous reviews of GenAI in education

The proliferation of GenAI has attracted considerable review articles in the literature (see summary in Table 1). For instance, Labadze et al. (2023) conducted a review of 67 articles, examining the benefits and challenges of AI chatbots in education. While emphasising benefits such as immediate support and personalised learning experiences, the authors also uncovered challenges related to reliability, accuracy and ethical considerations. Similarly, Ipek et al. (2023) conducted a systematic review to explore the positive and negative implications of integrating ChatGPT into educational settings. Their thematic analysis revealed potential benefits of ChatGPT integration such as translation and paraphrasing, summarising large volumes of texts, grading and assessment, personalised learning and data analysis. On the other hand, the authors reported concerns such as cheating, generating incorrect answers and creating bias. Additionally, Montenegro-Rueda et al. (2023) conducted a systematic review analysing the impact of ChatGPT in education. Their results indicate that ChatGPT implementation positively impacts the teaching-learning process in education. However, they also reported the importance of teacher training to effectively utilise the tool.

TABLE 1 Summary of previous reviews on GenAI in education.

Author	Type of review	# Included articles	Focus	Summary of main findings
Labadze et al. (2023)	Systematic review	67	To understand the full benefits of AI chatbots in education	(a) Students derive significant benefits from AI-powered chatbots across three principal domains: homework and study assistance, a personalised learning experience, and the development of various skills (b) For educators, the main advantages are the time-saving assistance and improved pedagogy (c) Challenges and concerns include reliability, accuracy and ethical considerations
Ipek et al. (2023)	Systematic review	40	To reveal the potential effects of ChatGPT on education	(a) GenAI performs high cognitive skills such as production, analysis and synthesis of information (b) However, creative measures should be enforced to avoid the negative aspects of AI
Memarian and Doleck (2023)	Literature review	63	Examines where ChatGPT is employed in educational literature and areas of potential, challenges, and future work	(a) Most studies explore ChatGPT through a commentary and non-empirical lens (b) Potentials of GenAI include personalised learning, assessments, feedback, etc. (c) Concerns include academic integrity
Ansari et al. (2023)	Scoping review	69	To understand how ChatGPT is being used in higher education across the globe	(a) GenAI serves as a convenient tool to assist teachers, students and researchers in various tasks (b) There are concerns related to accuracy, reliability, academic integrity and potential negative effects on cognitive and social development

Pradana et al. (2023)	Literature review and bibliometric analysis	74	To provide a review of existing research on the use of OpenAI ChatGPT in education	(a) Research on GenAI has attracted large number of publications across different countries (b) GenAI has the potential to revolutionise education, including its ability to conduct tasks that require creative intelligence
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TABLE 1 (Continued)

Author	Type of review	# Included articles	Focus	Summary of main findings
Kim et al. (2023)	Scoping review	33	To understand the current state of Large Language Model (LLM) use in medicine and to provide a guideline for future utilisation in academia.	(a) ChatGPT/LLM cannot be cited as an author in scientific manuscripts (b) Information produced by GenAI should be verified through human intuition (c) Use of GenAI in scholarly research should be documented
Vargas-Murillo et al. (2023)	Systematic review	16	Examines the impact of ChatGPT usage in education	(a) GenAI has the potential to offer educational opportunities to students (b) It is crucial to remember that the implementation of AI should be overseen by educators to improve academic integrity
Weidener and Fischer (2023)	Scoping review	12	To synthesise the literature on teaching AI ethics as part of medical education	(a) All extracted publications recommended teaching AI ethics in medical curricula due to the potential of AI in medicine (b) There are anticipated ethical challenges, including bias
Montenegro-Rueda et al. (2023)	Systematic review	12	To analyse the impact of the application of the ChatGPT tool in education	Although ChatGPT can enhance the educational experience, its successful implementation requires teachers to be familiar with its operation
Wang et al. (2024)	Systematic review	27	To explore the utility and availability from the technological affordances perspective	Findings revealed four affordances (accessibility, personalisation, automation and interactivity) and five challenges (academic integrity risk, response errors and bias, over-dependence risk, the widening digital divide, and privacy and security).

The study by Kim et al. (2023) explores a slightly different dimension by reviewing the ethical standards for using GenAI in academic medicine. They developed five key guidelines, emphasising that GenAI (a) cannot be cited as an author in scientific manuscripts, (b) should not be used to produce entire text in manuscripts without human intervention, and (c) may be utilised for editing and refining text. Additional guidelines include (d) the need for a basic understanding of the technology, and (e) the need for transparency, including documentation of its usage in scientific manuscripts. Their conclusion emphasises the necessity for authors to conscientiously consider the potential ramifications of their academic endeavours in healthcare and to consistently maintain the utmost ethical standards and integrity when employing GenAI.

Wang et al. (2024) conducted a systematic review of 27 publications to explore the utility and availability from the technological affordances perspective along with challenges surrounding the implementation of the technology. The findings of the study revealed four technological affordances (i.e., accessibility, personalisation, automation and interactivity) and five challenges (i.e., academic integrity risk, response errors and bias, over-dependence risk, the widening digital divide, and privacy and security). The authors recommend the integration of AI in education but advocate for the formulation of ethical guidelines and responsible use.

These reviews imply that GenAI can revolutionise education by enhancing various aspects such as personalised learning, efficient grading and content summarisation. However, they also highlight several concerns that need to be addressed. Ethical issues, such as bias in AI algorithms, privacy concerns regarding data usage, and the need for transparent AI operations, are significant challenges. Effective integration of GenAI in educational settings is, therefore, required to ensure that the technology is used responsibly and beneficially.

Review of search strategies used in previous reviews

Although previous reviews have made significant findings regarding the benefits and challenges of GenAI in education, a closer examination of their search strategies reveals both strengths and areas for improvement (see Table 2). Generally, these reviews searched a range of databases, including popular ones like Google Scholar, Scopus, Web of Science, IEEE Xplore, and domain-specific libraries such as Eric and MEDLINE. The inclusion criteria varied, with most reviews focusing on peer-reviewed articles published in English and within specific recent time frames. Some reviews also emphasised the need for studies with scientific methodologies or those directly related to ChatGPT in educational contexts.

However, we observed some weaknesses in the search strategies. First, there is inconsistent comprehensiveness of the search strings. Several reviews did not explicitly detail their search strategy, making it difficult to evaluate the replicability of their searches. This lack of detailed search strategies potentially limits the breadth of literature covered and may lead to the omission of relevant studies. Secondly, while some reviews applied stringent inclusion criteria to maintain high standards of evidence, these criteria narrowed the scope too much, excluding valuable interdisciplinary or grey literature that might offer broader insights. In response to the imperative for rigorous research, quality and transparency in review articles as proposed by Bond et al. (2024), our present study advanced beyond conventional review protocol by incorporating more expansive search strings and adopting flexible inclusion criteria. This approach aims to ensure a comprehensive exploration of the literature on GenAI across various educational disciplines.

TABLE 2 Search strategies of previous reviews.

Author	Search strings	Database used	Inclusion criteria
Ipek	Not explicitly reported	Eric, Wiley online library, Springer, Sage journals, Taylor & Francis, Google scholar, JSTOR, Science Direct	Research with scientific methodology; articles relating to ChatGPT in education; published between December 2022 and February 2023; published in English
Labadze et al. (2023)	("Education" or "Learning" or "Teaching") and ("Chatbot" or "Artificial intelligence" or "AI" or "ChatGPT")	ACM Digital Library, Scopus, IEEE Xplore, and Google Scholar	Articles published in peer review journals, books and book chapters; studies on AI in education; published 2018–2023; published in English
Memarian and Doleck (2023)	ChatGPT AND (education OR teaching OR learning OR pedagogy)	Web of Science, Scopus	Studies exploring ChatGPT in education
Ansari et al. (2023)	('ChatGPT' and 'Education')	Google Scholar, Taylor and Francis, Emerald, Sage, Elsevier, Science Direct, and PubMed	All types of peer-reviewed articles; published in English; published between 2022 and 2023; ChatGPT in higher education
Kim et al. (2023)	"artificial intelligence", or "machine learning", "natural language processing", "generative pre trained transformer", "ChatGPT", and "large language model"	MEDLINE	Articles exploring LLM; No restrictions to language or date of publication
Vargas-Murillo et al. (2023)	("ChatGPT") AND ("education" OR "learning" OR "e-learning")	Scopus, ScienceDirect, ProQuest, IEEE Xplore and ACM Digital Library	Articles written in English; published in the last 5 years; mentioned ChatGPT; related to the educational field; related to the learning process, teaching methodologies or different types of learning; from a scientific or specialised journal and/or magazine
Montenegro-Rueda et al. (2023)	("ChatGPT" OR "Chatbot" OR "Artificial Intelligence") AND ("Education" OR "Educational Institution")	Web of Science, Scopus, and Google Scholar	Published between 2022 and June 2023; Scientific article published in peer-reviewed journal; only theoretical and empirical research; English or Spanish; Use of ChatGPT in education

Research questions

Against this background, this review seeks to answer the following questions:

1. What types of articles explored GenAI between November 2022 and April 2024?
2. Which education level received the most attention regarding academic research and discussions on GenAI?
3. Which education disciplines predominantly engaged in discussions and research activities of GenAI?
4. What research areas were predominantly explored in publications related to GenAI?
5. Which research areas were mostly explored by various education disciplines?
6. How do different education disciplines conceptualise the application of GenAI in the field of education?

METHODOLOGY

This review employs systematic mapping (SM) methodology, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 recommendations (Page et al., 2021, see checklist in OSF¹). This approach shares similarities with systematic reviews, albeit with noticeable distinctions. Unlike systematic reviews that focus on synthesising and appraising results, SM employs broader inclusion criteria with the specific aim of mapping out topics (Dicheva et al., 2015). This methodology provides a structured framework for categorising published reports and their respective findings (Bond et al., 2020; Yusuf & Noor, 2023). Knowledge generated through the application of SM proves instrumental for subsequent research endeavours and as a preliminary study preceding more in-depth systematic reviews (Kitchenham et al., 2011).

Literature search

For a comprehensive literature search, articles were extracted from three prominent databases: Web of Science, ScienceDirect and Scopus. The literature search was conducted on 25 April 2024 and encompasses published papers from 1 November 2022 to 30 April 2024.

The starting timeframe represents the emergence and ongoing development of generative AI in the academic environment (Ansari et al., 2023; Montenegro-Rueda et al., 2023) and the ending timeframe represents articles whose potential publication date is 2024. We searched the databases using a search string adapted from Bond et al. (2024); see Figure 1.

The first search string topic comprises keywords commonly employed by authors or indexed databases to reference GenAI technologies. In this section, we included prominent GenAI tools such as ChatGPT and Gemini (Yusuf et al., 2024) because some research titles, abstracts or keywords might not include the other terms. This approach was suggested by Yusuf and Noor (2023) in their recent mapping study. The second search string topic represents the educational context of the research. One of the databases (i.e., ScienceDirect) imposed a limitation on the maximum number of Boolean connectors (eight per field) that could be included in a search. For this reason, we refined our search strings and included eight Boolean connectors to meet the requirements of this database: ("generative artificial intelligence" OR "generative AI" OR genAI OR "large language model" OR ChatGPT) AND (education OR teaching OR learning OR students). The search terms yielded 3241 publications (see Figure 2), all of which underwent subsequent screening processes for further refinement and analysis.

GenAI	“generative artificial intelligence” OR “generative AI” OR “chatbot” OR “large language model” OR “conversational agents” OR “genAI” OR “chatGPT” OR “gemini”
AND	
Education	“higher education” OR college* OR undergrad* OR “graduate” OR “postgrad*” OR “K-12” OR “kindergarten*” OR “corporate training*” OR “professional training*” OR “primary school*” OR “middle school*” OR “high school*” OR “elementary school*” OR “vocational education” OR “adult education” OR “workplace learning” OR “corporate academy”

FIGURE 1 Search string.

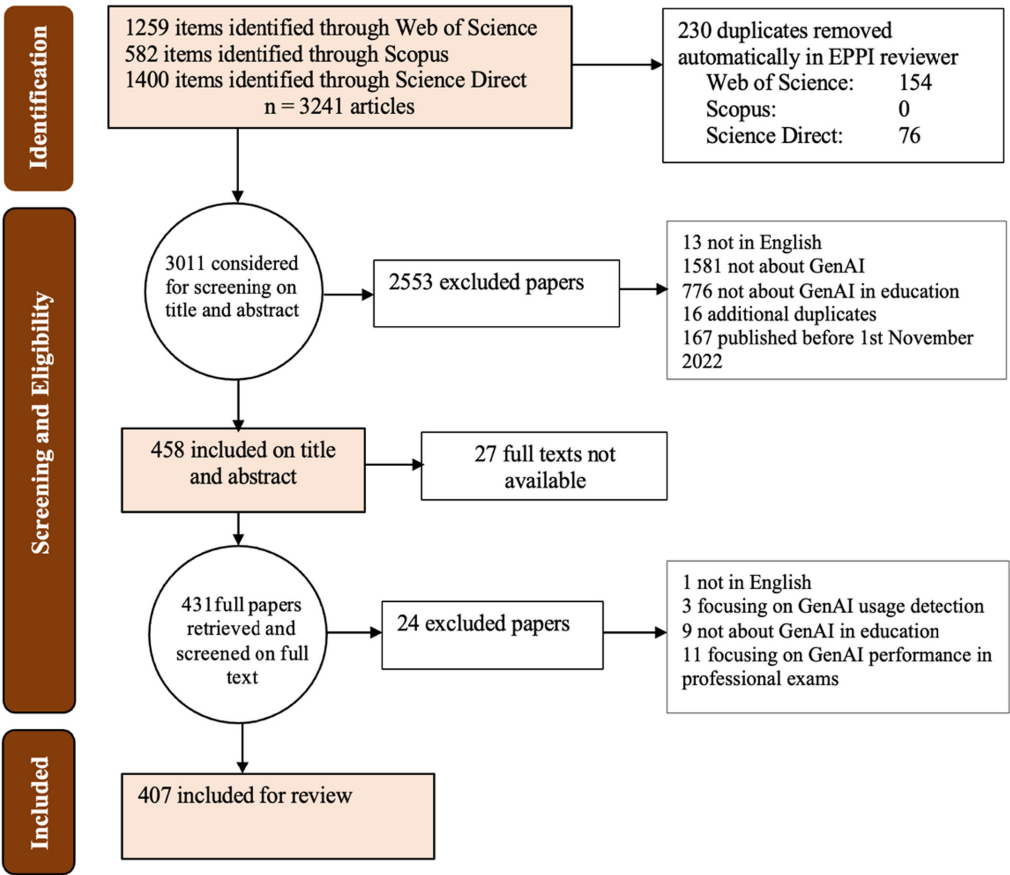


FIGURE 2 PRISMA 2020 flow diagram.

Inclusion/exclusion criteria and screening

Following the article search, all publications were exported as .ris files and imported into EPPI Reviewer Web version 6 (Thomas et al., 2023), a web tool for conducting systematic reviews. Upon importation of the articles, the web tool automatically removed 230 publications, of which the remaining 3011 underwent title and abstract screening. The screening process was guided by three criteria: (a) the use of English as the medium for reporting, (b) focusing on GenAI in education, and (c) published on or after 1 November 2022. It is

important to note that our selection process did not discriminate against any publication provided they met our eligibility criteria. Consequently, we included various forms of publications, such as original articles, reviews, commentaries, letters, posters, editorials and notes in our article extraction. However, reports unrelated to GenAI ($n=1581$) or not about GenAI in education ($n=776$) were excluded. As we proceeded to the screening, we found 16 additional duplicates not discovered by the EPPI Reviewer. Furthermore, we excluded 167 articles published before 1 November 2022. Overall, the initial screening produced 458 publications, which were retained for full-text screening.

The fundamental screening criteria for the full text involves publications reporting or discussing GenAI in education. However, in addition to the criteria employed at the title and abstract screening, we excluded publications (a) detecting the use of GenAI in scholarly writing (e.g., Jakesch et al., 2023; Nguyen et al., 2023), and (b) those reporting the performance of GenAI in professional examinations such as medical licence and MBA examinations (e.g., Ali et al., 2023; Allen & Woodnutt, 2023). The only performance of GenAI included in our study encompasses instances related to personalised learning and autoethnography (e.g., Araújo & Saúde, 2024; Stojanov, 2023). The decision to exclude instances of professional examinations was made to maintain a targeted and coherent scope within these particular domains. The assessment of GenAI performance in professional examinations only aimed to gauge its intelligence characteristics rather than its potential contributions or risks in the realms of learning and research.

As we further our screening, we were unable to retrieve full texts for 27 articles, which were subsequently excluded. We discovered that some publications appeared to satisfy our criteria in their titles and abstracts but actually focused on other domains. After the full-text screening, we excluded 24 articles based on our eligibility criteria, with 407 publications included in our database. A reference list of the extracted publications can be found in the Open Science Framework (OSF).ⁱⁱ To mitigate or minimise the issue of coding bias, two researchers screened the 3011 articles based on titles and abstracts in the EPPI Reviewer Web. After five rounds of comparison coding (average 550 items), this double coding yielded an almost perfect agreement (Cohen's Kappa $\kappa=0.81$). Following the reconciliation of six articles, another five rounds of comparison coding (average 91) were conducted by the same authors who screened the 431 full texts, reaching almost perfect agreement (Cohen's Kappa $\kappa=0.86$).

Data extraction

To map information relevant to the research questions, we employed a classification methodology similar to that utilised by Yusuf and Noor (2023). This approach involved manually coding all study features based on the frequency of occurrence of mapped variables. Given the potential limitations of relying solely on abstracts for comprehensive information, we took a thorough approach by reading significant sections of the full texts. This ensured that the information within the complete texts guided the mapping process. The data mapping process was manually done by two authors in the EPPI Reviewer Web, where articles were mapped based on (a) article type (e.g., original article, reviews, etc.), (b) publication type (e.g., journal, book chapter, conference), (c) discipline (e.g., Science education, medical education, etc.), (d) educational level (e.g., higher education, general education, K-12), (e) research area, and (f) GenAI conceptualisation.

To extract useful information concerning research areas and how GenAI is conceptualised, we employed content analysis, which involved systematic coding and thematic classification. Our analysis began with rigorous manual coding from the titles and abstracts, followed by a comprehensive examination of the full texts to identify recurring themes and patterns. This process of thematic analysis unfolded iteratively, incorporating multiple

TABLE 3 Summary of article mapping and extraction.

S/N	Category	Description	Method used
1	Article type	Entails the type of extracted publication, including original article, reviews, editorial, commentaries, etc.	Reading of titles and abstracts
2	Publication type	The type of publication, including journal, conference proceedings, book chapter, etc.	Reading of titles and abstracts
3	Discipline	The educational discipline that dominates a publication, including general education, medical education, programming education, etc.	Reading of titles, abstracts and full texts
4	Educational level	The level of education focused on by a publication, including higher education, general teaching and learning, elementary level, etc.	Reading of titles, abstracts and full texts
5	Research area	The focus of a publication, including perspectives and reactions, impact and implication, ethical implications and risk, etc.	Content analysis by reading of titles, abstracts and full texts
6	Conceptualisation	How GenAI application in education is conceptualised.	Content analysis by reading of titles, abstracts and full texts

rounds of refinement and discussion. Collaboration was integral to our methodology, as we sought validation from colleagues and experts in the field. Additionally, we conducted cross-verification of themes by revisiting a subset of publications, ensuring alignment with the details present in our dataset. Through this iterative process, we arrived at important discursive themes. These themes represent a consolidation of our dataset and a distillation of the diverse research areas and conceptualisations of GenAI application in education across a spectrum of disciplines.

A summary of the article mapping and extraction process is presented in [Table 3](#). After mapping and classifying the articles, the overall article report was extracted using the ‘quick question report’ function in the EPPI Reviewer Web and subsequently converted to an Excel file for data analysis (see [OSFⁱⁱⁱ](#)). Evidence of the complete article mapping and extraction codes can be found in the EPPI Visualizer online repository.^{iv}

Limitations

While conducting this systematic mapping review, it is important to acknowledge certain limitations that may impact the interpretation of our findings. One notable limitation is the absence of critical quality appraisal of the included studies. However, as this was a mapping review rather than a systematic review and its intent was to map the state of the field, a quality appraisal was not necessary (Grant & Booth, 2009). We exercised caution and diligence during the study selection process, adhering strictly to our predefined inclusion criteria. By adopting these criteria, we aimed to minimise the risk of including irrelevant studies in our review. Additionally, by transparently documenting our search strategy and selection process, we aimed to enhance the transparency and reproducibility of our findings. Despite this, the absence of critical quality appraisal may introduce uncertainty regarding the reliability and robustness of the synthesised evidence. Readers should interpret the findings of this review with this limitation in mind and exercise caution when drawing conclusions or making recommendations based solely on the included studies.

Analysis

We conducted all analyses requiring basic frequencies and cross-tabulations using both EPPI Reviewer and Excel. The frequency analysis includes plotting a bar chart that depicts frequency distribution (RQ1–RQ4) and the cross-tabulations involve identifying patterns and associations between variables (RQ5 and RQ6). To provide visual overviews and publicly accessible resources beyond the scope of this article, we created an EPPI Visualizer database using the EPPI Reviewer tool. We started by clicking the ‘Setup Visualizations’ link on the EPPI Reviewer user homepage. Following the on-screen prompts, we added our ‘data extraction’ code containing all the final data categories. We then created ‘evidence and gap maps’ for more interactive visualisation. Previous versions of the EPPI Reviewer required importing a JSON report of all included studies into the EPPI Mapper application (Thomas et al., 2023). Although this approach can produce a downloadable, standalone visual, it is sometimes browser-dependent and requires specialised expertise. With the emergence of recent versions of EPPI Reviewer,^v we employed a more straightforward approach by clicking the ‘create map’ button in the EPPI Visualizer. Although the evidence and gap maps created through this method require web access, they offer more standardised viewing across various platforms. Our EPPI Visualizer database can be accessed online.^{vi}

RESULTS

RQ1: What types of articles were published between November 2022 and April 2024?

While only three studies (0.7%) were published in 2022 (Guo et al., 2022; Li et al., 2022; Rahim et al., 2022), 234 (57.5%) were published in 2023 and 170 (41.8%) had already been published in the first 4 months of 2024 (see Figure 3). Of the 407 publications extracted for this study, 323 (79.4%) are original research articles conveying empirical research findings (e.g., Abbas et al., 2024; Yusuf et al., 2024) or theoretical articles (e.g., Glaser, 2023; Lim et al., 2023). Others included 33 (8.1%) reviews (e.g., Ansari et al., 2023; Kim, 2023), 20 (4.9%) opinion papers (e.g., Castonguay et al., 2023; Grassini, 2023), 10 editorials (e.g., Hwang & Chen, 2023; Mendiola, 2024), nine (2.2%) commentaries (e.g., Beck & Levine, 2023; Dianova & Schultz, 2023; Eager & Brunton, 2023), seven (1.7%) letters (Arachchige, 2023; Currie et al., 2023), and five (1.2%) discussions (Bender, 2024; Gosak et al., 2024). More information can be found in Appendix A.

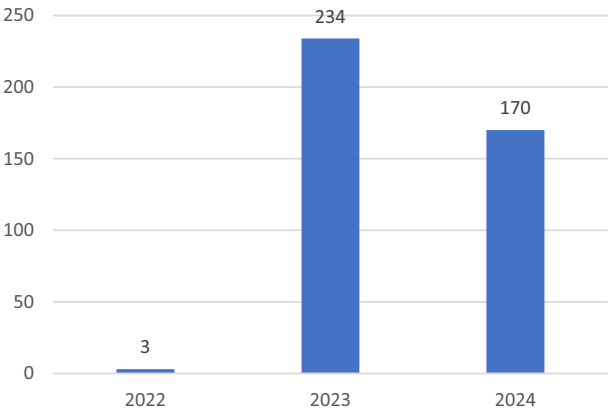


FIGURE 3 Distribution of publications by year.

RQ2: Which educational level received the most attention regarding academic research and discussions on GenAI?

The analysis revealed the educational level that received the most attention in GenAI research (see [Figure 4](#)). A significant portion of publications ($n=306$, 75.2%) centred on higher education contexts (e.g., [Abbas et al., 2024](#); [Abdaljaleel et al., 2024](#)). Following closely was the attention given to general education ($n=88$, 21.6%, e.g., [Steele, 2023](#); [Zirar, 2023](#)), while K-12 education ($n=13$, 3.2%, e.g., [Baha et al., 2023](#); [Su & Yang, 2023](#)) received fewer discussions. More information can be found in [Appendix B](#). Here, 'higher education' is defined as postsecondary institutions that typically confer degrees, diplomas or certificates upon completion of a course of study ([Teague, 2023](#), p. 5). Meanwhile, 'general education' encompasses a broad form of teaching and learning.

Articles falling into this category either lacked specificity regarding their educational focus or engaged in discussions that spanned a wide range within the educational community. K-12 education encompasses the years of formal schooling starting from kindergarten (age 5 or 6) up to the completion of high school or the 12th grade (age 17 or 18).

RQ3: Which education disciplines predominantly received discussions on GenAI?

Our analysis revealed 28 educational disciplines receiving attention about the application of GenAI (see [Figure 5](#)). General education ($n=220$, 54.1%) was the focus of most academic discussions on GenAI (e.g., [Chiu, 2023](#); [Xu et al., 2024](#)). This is followed by medical education ($n=54$, 13.3%, e.g., [Totlis et al., 2023](#); [Wu et al., 2024](#)), language education ($n=32$, 7.9%, e.g., [Barrot, 2023](#); [Boudouaia et al., 2024](#)), computing and engineering ($n=21$, 5.2%, e.g., [Sun et al., 2024](#); [Yilmaz & Yilmaz, 2023](#)), and nursing education ($n=13$, 3.2%, e.g., [Tam et al., 2023](#); [Yalcinkaya & Yucel, 2024](#)). The 'others' category ($n=17$, 4.2%) includes less prominent disciplines such as nutrition and dietetics education ([Barker et al., 2024](#)), sport management education ([Keiper et al., 2023](#)), and social media education ([Kim, 2023](#)). More information can be found in [Appendix C](#).

Our synthesis shows that GenAI offers educational potential across disciplines amid concerns. For example, in the field of language education, [Boudouaia et al. \(2024\)](#) investigated

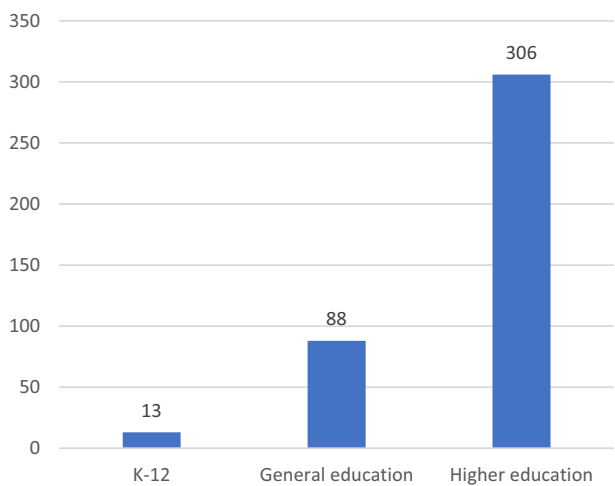


FIGURE 4 Article distribution by educational level.

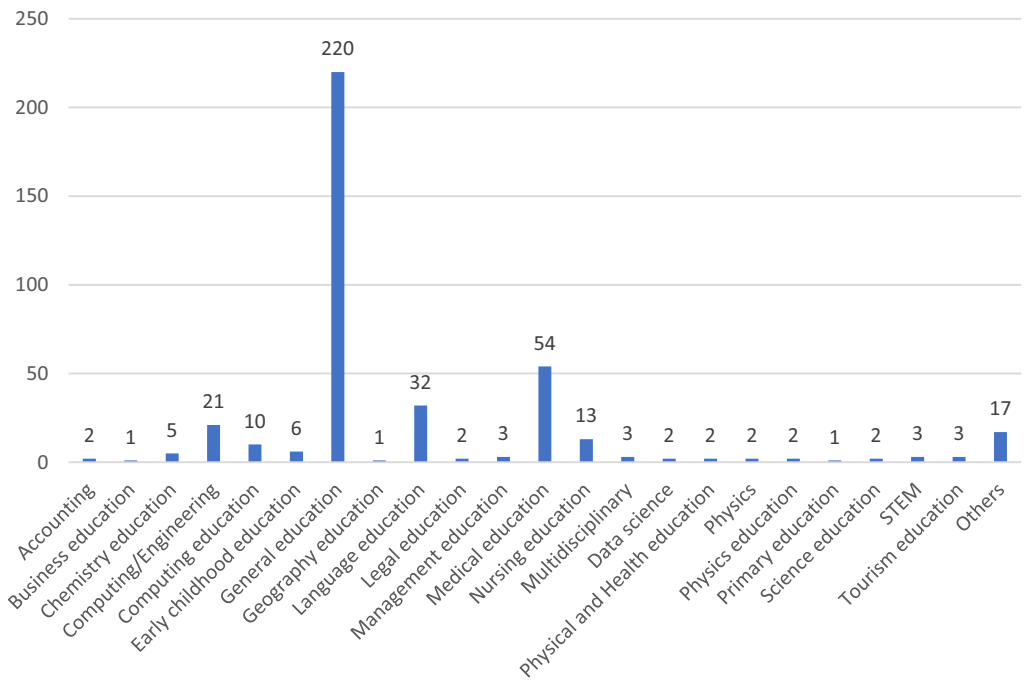


FIGURE 5 Article distribution by discipline.

the use and acceptance of ChatGPT-4 in English as a foreign language (EFL) writing among undergraduate students in Algeria. The findings revealed that students exposed to ChatGPT significantly outperformed those exposed to a control condition in terms of their EFL writing skills. Additionally, there were notable improvements in their perceived usefulness, perceived ease of use, attitudes and behavioural intention toward ChatGPT-4.

In the context of programming education, Haindl and Weinberger (2024) examined the use of ChatGPT by part-time undergraduate students in a 5-week Java programming course. The study involved collecting anonymous feedback from students after each exercise, where they rated various aspects of ChatGPT's suitability. Although the majority of students viewed ChatGPT positively for learning programming concepts, its suitability for specific implementation tasks received mixed reviews. A substantial number of students found it easy to adapt ChatGPT-generated code for implementation tasks, and they primarily used it to acquire background knowledge, learn syntax, and learn programming concepts. However, some students abstained from using it due to concerns about not gaining sufficient programming proficiency, encountering partially incorrect or misleading code, preferring an independent working style, or scepticism about its benefits. These results call for continued research to optimise the integration of GenAI in education, ensuring it meets the unique demands of each field.

RQ4: Which research areas were predominantly explored in publications related to GenAI?

As we continued to explore the GenAI literature, we found a wide range of research focus across the extracted studies. The content analysis revealed 78 keywords which were subsequently framed under eight thematic areas (see Figure 6, Appendix D). In this analysis, it is important to note that article classification is not mutually exclusive. Therefore, an article might be classified under more than one research focus. A large proportion of the publications

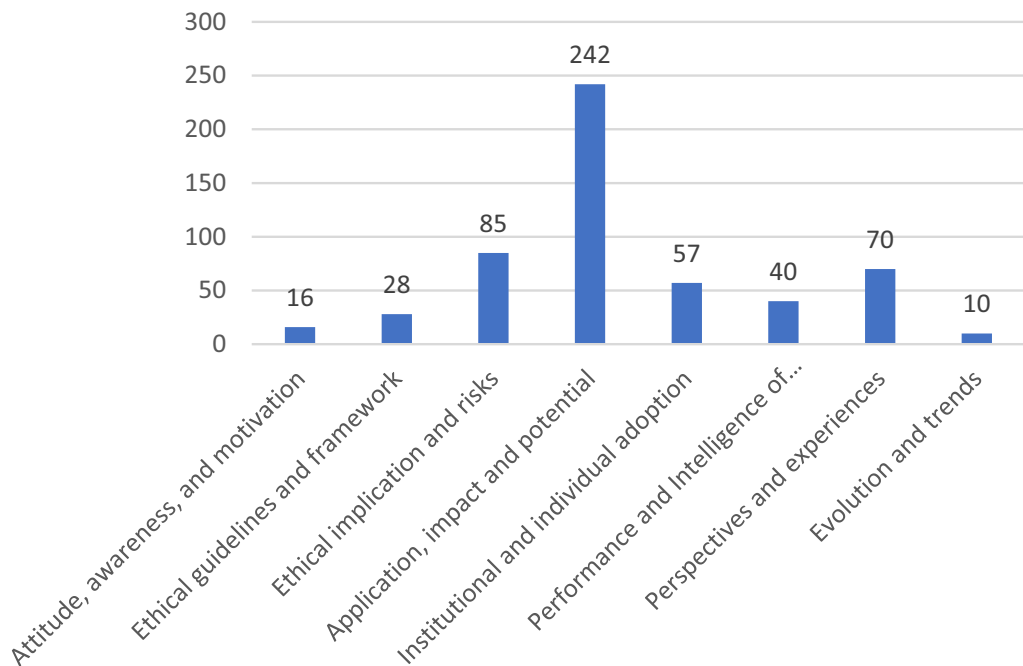


FIGURE 6 Article distribution by research areas.

($n=242$, 59.5%) explored the impact and potential of GenAI. This is followed by publications that explored ethical implications and risks ($n=85$, 20.9%), perspectives and experiences ($n=70$, 17.2%), and institutional and individual adoption ($n=57$, 14%). Others include performance and intelligence ($n=40$, 9.8%), ethical guidelines and framework ($n=28$, 6.9%), attitude, awareness and motivation ($n=16$, 3.9%), and evolution and trends ($n=10$, 2.5%).

Application, impact and potential

Studies categorised under the theme of ‘application, impact, and potential’ demonstrate a keen understanding of the far-reaching implications of GenAI within the academic environment. These studies examined how the integration of GenAI stands to influence and shape academic practices, research methodologies and educational outcomes. Studies under this thematic category employ various research approaches to examine these impacts and potentials. For example, Yilmaz and Yilmaz (2023) employed an experimental design to investigate how integrating ChatGPT into programming education enhances students’ computational thinking skills, programming self-efficacy and motivation. Their findings revealed significant improvements in the experimental group’s computational thinking skills, programming self-efficacy and motivation compared to the control group.

Similarly, Sun et al. (2024) utilised a mixed-method design to investigate the impact of ChatGPT-facilitated programming (CFP) on college students’ programming behaviours, performances and perceptions. Their findings revealed that students in the CFP mode demonstrated increased engagement with debugging, error messages and feedback interactions. Students’ interview data also highlighted significant improvements related to ChatGPT’s usability and stages of usage. Lastly, perceptions toward ChatGPT improved post-CFP practice, affecting perceived usefulness, ease of use and intention to use. In another study, Thorat et al. (2024) employed a systematic review protocol to explore ChatGPT’s role in

dental education. Their study highlights several impacts of ChatGPT across various educational facets, including personalised learning pathways, integration into virtual patient simulations, continuous accessibility, multilingual support, interactive dental dictionary functionality, evidence-based learning and real-time assessment of students.

Ethical implications and risks

Publications within this theme focused on the ethical implications of using GenAI in academia. These studies dissect various threats posed by GenAI, including concerns related to privacy, transparency, algorithmic bias and the potential for reduced critical thinking. For instance, Castonguay et al. (2023) reflect on the disruptive impacts of ChatGPT and found it has a tendency to simulate and produce false information, which undermines academic integrity. Similarly, Chauncey and McKenna (2023) investigated the ethical issues surrounding the use of AI chatbots in education and identified issues such as misinformation, plagiarism, difficulties in citing sources, identity theft, AI cloning, mind reading, prompt injection and AI hallucination. Zhang, Amos, and Pentina (2024) added another layer to these ethical concerns by discussing the potential for ChatGPT-generated content to facilitate academic misconduct. They find that the low detectability of ChatGPT-based plagiarism undermines traditional rational choice theory, which relies on the perceived costs and benefits of cheating. Their study reveals that moral disengagement is a significant predictor of ChatGPT-based plagiarism, along with perceived benefits, punishment severity and informal sanctions.

These ethical issues highlight the limited capabilities of AI for independent and autonomous reasoning (Osasona et al., 2024), reinforcing the need for vigilant and ethical use of such technologies in educational settings. Given these ethical concerns, studies within this theme emphasised the importance of responsible use of AI tools, adherence to academic integrity guidelines, and addressing concerns related to data privacy, bias, reliability and transparency. They also emphasise the importance of nurturing human skills like empathy, critical thinking and problem-solving in educational curricula, as these cannot be replaced by AI.

Perspectives and experiences

Publications that fall within this theme largely collected students' and educators' opinions concerning the implication of GenAI (e.g., Chan & Hu, 2023; Yusuf et al., 2024) or reported the potential of GenAI from a personal experience (e.g., Haindl & Weinberger, 2024). Many of these studies employed instruments such as surveys, interviews and autoethnographic studies, although further research could explore data collection and analysis methods in greater detail. For example, Chan and Hu (2023) conducted a survey to explore university students' perceptions of GenAI technologies in higher education. The study found considerable students' familiarity with ChatGPT, their willingness to engage, potential benefits and challenges, and effective integration. In addition, Barrett and Pack (2023) surveyed 68 educators and 158 students to compare their perceptions on the appropriate use of GenAI in the writing process. Their results indicate minor disagreements between students and teachers on the acceptable use of GenAI tools in the writing process, as well as a lack of preparedness for GenAI at both the classroom and institutional levels. Furthermore, Yusuf et al. (2024) surveyed 1217 participants across 76 countries to understand the potential of GenAI from a multicultural viewpoint. The authors highlighted several perceived benefits

and concerns of GenAI but stressed that these perceptions could be influenced by cultural norms.

Institutional and individual adoption

This theme reflects the growing interest and involvement of both organisations and individuals in adopting GenAI technologies into their practices. Studies within this theme explored the dynamics of GenAI adoption. One similarity among these studies centres on the factors responsible for GenAI usage and adoption. Several factors were reported in these studies. For example, Tiwari et al. (2023) identified perceived usefulness, social presence, legitimacy of the tool, and enjoyment and motivation as the major factors. Similarly, Strzelecki (2023) identified behavioural intention and personal innovativeness as the dominant determinants of the usage behaviour of GenAI. However, the author identified 'habit' as the best predictor of behavioural intention to use GenAI, followed by performance expectancy and hedonic motivation. In the same position, Tian et al. (2024) found that users' perceived confirmation significantly influences their satisfaction and subsequent intention to continue using AI Chatbots. However, 'personal innovativeness' emerges as a critical determinant shaping users' behavioural intentions.

Although these factors were widely reported, existing literature suggests that usage intention may vary by teacher and student generation. For instance, Chan and Lee (2023) found that Generation Z participants were optimistic about the potential benefits of GenAI, anticipating improvements in productivity, efficiency and personalised learning, and expressed intentions to use GenAI across diverse educational contexts. Conversely, participants from Generation X and Y acknowledged GenAI's benefits but expressed greater concerns about over-reliance and ethical and pedagogical implications. Overall, these studies highlight the need to adopt GenAI while addressing the potential challenges educational institutions may face.

Performance and intelligence

This theme encompasses studies assessing the performance of GenAI systems in specific educational tasks and their intelligence in assisting learners and educators. Tasks range from answering questions and providing study aids to evaluating assignments and facilitating learning activities. Despite variations in educational domains and tasks evaluated, these studies collectively explored the capabilities of GenAI models using performance metrics. For instance, Araj and Brooks (2024) evaluated the performance of ChatGPT as a study aid and found that it is comparable to search engines in assisting medical students during surgery clerkships. Similarly, Kavadella et al. (2024) reported significant improvements in knowledge examination scores among dental students using ChatGPT for learning assignments.

In addition, Parker et al. (2023) explored ChatGPT's capability for automated writing evaluation in nursing education and found its utility in providing feedback aligned with assessment criteria. Lu et al. (2024) contribute to the discussion by investigating how ChatGPT can effectively complement teacher assessment of students' academic writing. The study revealed that ChatGPT efficiently scored students' writing, showing moderate to good consistency with teacher evaluations. However, the authors observed significant differences in feedback quantity and types between ChatGPT and teacher assessments. From this synthesis, it becomes evident that GenAI tools are, to an extent, intelligent to perform complex educational tasks.

Ethical guidelines and framework

Publications falling within 'ethical guidelines and framework' acknowledge the ethical concerns of integrating GenAI in education. These studies either assess existing guidelines across institutions or propose new guidelines for GenAI integration. For example, assessments of institutional frameworks indicate weak policies guiding GenAI usage (see De Maio, 2024; Luo, 2024). Regarding framework proposals, Chauncey and McKenna (2023) formulated a conceptual framework for the responsible use of GenAI. For educators, key steps include prompt design, response evaluation, constructed response questions and challenging assignments. For students, crucial steps include prompt design, response evaluation, creative ideation and innovative application. Similarly, Kim et al. (2023) proposed guideline statements for GenAI usage, emphasising the need for (a) excluding GenAI as an author in scientific manuscripts, (b) human insight when GenAI is used for academic writing, and (c) documenting areas where GenAI is used in scholarly manuscripts. Overall, these studies investigate the ethical dilemmas and challenges posed by AI systems, proposing and assessing frameworks that aim to guide developers, practitioners and policy makers in navigating the complex landscape of ethical considerations.

Attitude, awareness and motivation

This theme encompassed examinations of attitudes toward employing GenAI, assessments of students' awareness, exploration of its facilitation of cultural awareness, investigations into the role of motivation in GenAI utilisation, and explorations of how GenAI motivates students to engage in learning. For example, Sallam et al. (2023) developed and validated a survey instrument to gauge healthcare students' attitudes toward ChatGPT. They found a generally positive attitude toward using ChatGPT among students. However, they noted that factors such as perceived risks, attitudes toward technology, social influence and potential anxiety provoked by ChatGPT could limit this positivity.

Zhang, Yang, and Tong (2024) employed a person-centred approach to uncover students' attitudinal profiles regarding their usage of ChatGPT. Their study revealed three profiles: positive embracers, cautious rational optimism and a complex attitudinal profile. Among the participants, 80.2% exhibited cautious rational optimism, while positive embracers showed higher attitudes toward using ChatGPT. Furthermore, Zheng and Stewart (2024) investigated the potential of ChatGPT in improving EFL cultural awareness through reframing the moral dilemmatic stories. Their findings indicate that ChatGPT supports immersive language learning by increasing exposure to authentic resources that facilitate language acquisition. Overall, these studied attitudinal, motivational and cultural awareness benefits of GenAI technologies, while also highlighting the importance of addressing concerns and maximising its pedagogical advantages in diverse educational settings.

Evolution and trends

The studies within this theme offer insights into the trajectory and current trends of GenAI technologies. These studies primarily employed bibliometric analysis, with some original articles discussing its evolution. The central focus is on investigating the predominant applications of GenAI and the scientometric evaluation of previous studies. One of the prominent researches was the one conducted by Bahroun et al. (2023) to investigate the diverse applications of GenAI in education. Part of the study's findings highlighted GenAI use in enhancing translation accuracy, improving patient care, fostering critical thinking skills, empowering

students to generate innovative solutions, and enhancing the teaching and learning of programming concepts.

Similarly, Liu et al. (2024) conducted a bibliometric analysis to provide an overview of the current development of GenAI in education. Their findings indicated a rapid growth in the number of annual research papers on GenAI in education. Through co-citation analysis, the authors discovered that the ten most cited articles in this field were all published from 2018 onwards, with six articles specifically focusing on the improvement and optimisation of language models and training methods. Zheltukhina et al. (2024) also conducted a bibliometric analysis to explore research trends in GenAI. Based on co-occurrence keywords, their study identified five key research areas: the potential utility of GenAI to facilitate nursing education, ethical implications and integrity issues, the impact of GenAI on various learning outcomes, its application in driving medical education, and its numerous potential uses attracting research interest from broader perspectives. In summary, findings from this thematic category inform future research directions and practical implementations.

RQ5: Which research areas were mostly explored by different education disciplines?

We conducted a cross-tabulation to examine the association between disciplines and the research focus of the extracted publications (see Figure 7, see also Appendix E). Our analysis revealed some important patterns. Discussions concerning ‘ethical implications and risks’ prominently featured disciplines such as medical education (e.g., Armitage, 2023; Feng & Shen, 2023), nursing education (e.g., Gosak et al., 2024; Tam et al., 2023), and general

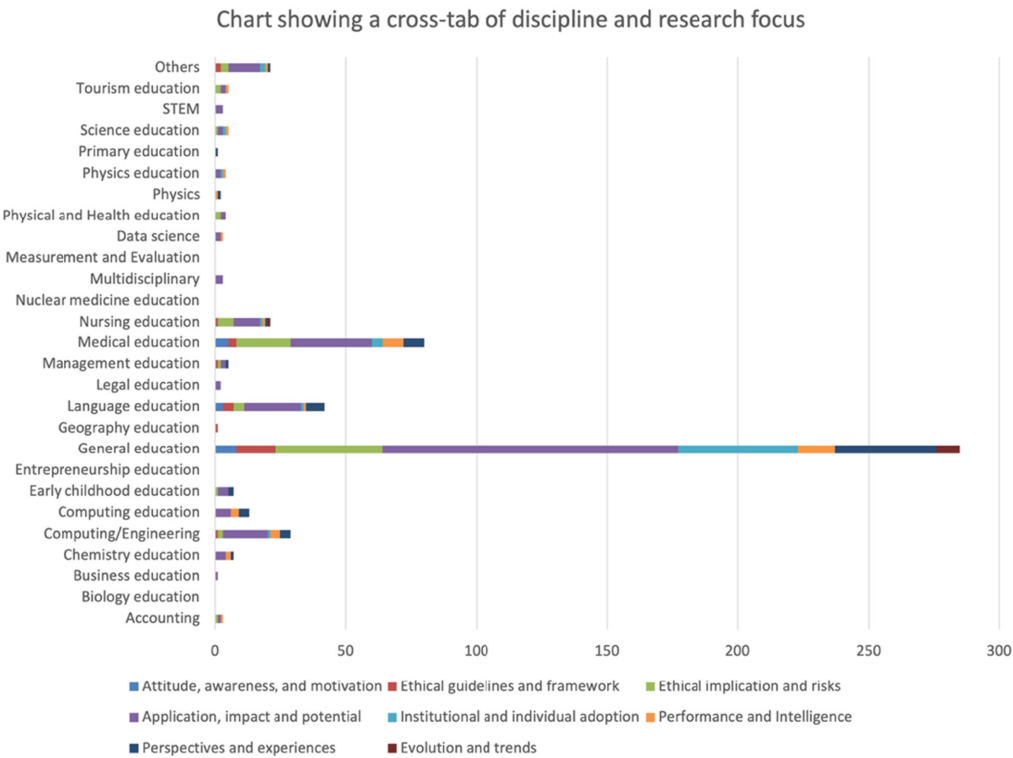


FIGURE 7 Research areas explored by various education disciplines.

education (e.g., Kirwan, 2023; Perkins, 2023). Similarly, the focus on ‘performance and intelligence’ was more pronounced in disciplines such as computing (Joshi et al., 2024), medical education (e.g., Araj & Brooks, 2024), and general education (e.g., Ahmad et al., 2023). However, the focus on ‘application, impact, and potential’ was pronounced across all disciplines. Based on these patterns, we argue that while there is a broader discourse on GenAI in education, certain disciplines exhibit specialised interests, likely influenced by the ethical considerations inherent to the field and the practical implications of AI within that specific domain. An interactive evidence gap map supporting this analysis can be found in the ‘Evidence and Gap Maps’ section of the EPPI Visualizer platform.^{vii}

RQ6: How is the application of GenAI in education conceptualised by different education disciplines?

We extended our analysis with a comprehensive synthesis to understand how different education disciplines conceptualise the application of GenAI in education. Our synthesis (see Figure 8) revealed five thematic constructs (see Appendix F), emerging through a cross-tabulation (see Appendix G). An interactive evidence gap map supporting this analysis can be found in the ‘Evidence and Gap Maps’ section of the EPPI Visualization web tool.^{viii}

Theme 1: A tool for pedagogical enhancement

The overarching conceptualisation of GenAI in this theme involves leveraging the tools to augment and refine pedagogical practices across various education disciplines. This theme

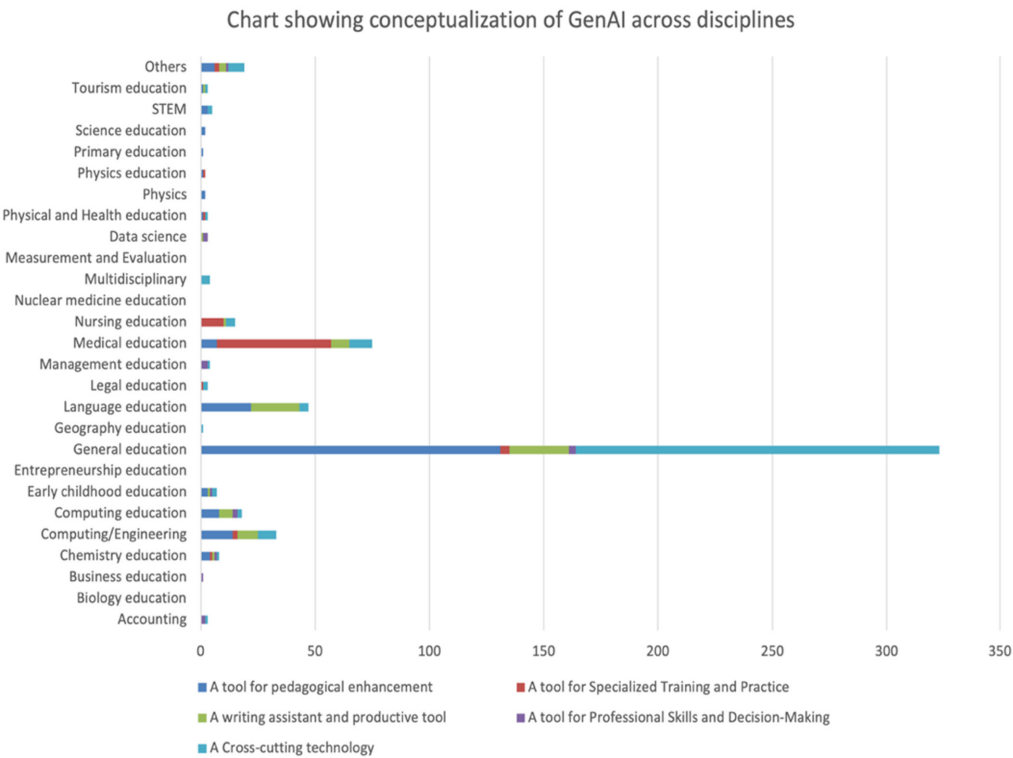


FIGURE 8 Map showing how various academic disciplines apply GenAI in education.

highlights the transformative potential of GenAI in reshaping how educators deliver content and engage with learners. A focus on personalised learning experiences, adaptive teaching methodologies and interactive educational content characterises the application of GenAI within this theme. Publications dominating this theme emanate from education disciplines classified as 'General Education', 'Programming Education', 'Science Education', STEM and 'Language Education'.

Within the 'General Education' context, the focus was on adapting the tools to analyse individual learning or collect wider opinions to understand how the tools improve learning. Chan and Hu's (2023) study reveals a positive student reception to GenAI in teaching and learning, emphasising personalised support, writing assistance and research capabilities. Lee et al. (2023) contributed by discussing the design considerations for sustaining student discourse and knowledge creation through ChatGPT architecture, showcasing its potential for pedagogical enrichment. Tlili et al.'s (2023) examination of ChatGPT adoption provides further insights into how GenAI tools can enhance pedagogical practices, addressing concerns and emphasising responsible integration.

Within the context of 'Programming Education', the focus is on applying GenAI tools to enhance programming skills and learning outcomes. Reflecting on this theme, Bucaioni et al. (2024) scrutinised the capability of ChatGPT to assist in coding assignments. Their experimental results showed that while ChatGPT is effective in helping students solve easy and medium programming problems, it struggles with more complex tasks. On the other hand, Yilmaz and Yilmaz (2023) investigated the impact of ChatGPT on programming learning outcomes. Their findings revealed that students exposed to ChatGPT exhibited higher computational thinking skills, programming self-efficacy and motivation for learning programming compared to their control counterparts. In a follow-up study, the authors confirmed that GenAI improved thinking skills, facilitated debugging and increased self-confidence. Despite these benefits, the authors expressed reservations, including concerns about fostering laziness, generating incorrect responses and causing professional anxiety in students.

Incorporating GenAI into studies in science education is driven by the aim to simulate experiments, analyse data and construct dynamic models. At a broader level, researchers employ GenAI to unravel insights into how students' learning experiences can be optimised. For example, Cooper (2023) investigates ChatGPT's responses to science education queries, potential utilisation in science pedagogy, and reflections on its role as a research tool. However, the study does not shy away from raising ethical concerns, underscoring the imperative of responsible use and the cultivation of critical thinking. Meanwhile, in language education, authors such as Yuan (2023) confront the controversy surrounding ChatGPT's presence in education, specifically emphasising its potential benefits for English language teaching and learning. The articles delve into practical experiments showcasing how GenAI can foster EFL and L2 classrooms. Yet, inherent in this exploration are critical questions surrounding the ethical dimensions and potential pitfalls associated with integrating GenAI into educational practices.

Theme 2: A tool for specialised training and practice

This theme revolves around the application of GenAI to cater to the unique needs of specialised education disciplines. The theme acknowledges the capacity of GenAI to simulate real-world scenarios, enhance decision-making skills, and bridge the gap between theoretical knowledge and practical application. The application of GenAI within this discourse is dominated by disciplines such as 'Medical Education' and 'Nursing Education'. In the context of medical education, GenAI is harnessed for virtual patient simulations, diagnostic support

and case writing. Authors in this discipline recommend the technology for realistic scenarios, such as helping students prepare medical cases and diagnostic reports. For example, Totlis et al. (2023) conducted an autoethnographic study where they had a conversation with ChatGPT to assess its capabilities in providing accurate and comprehensive anatomical descriptions, clinical relevance, and relationships between structures. The results demonstrate ChatGPT's ability to generate well-structured anatomical descriptions, offer concise summaries, and even create quizzes and matching questions.

Nursing education utilises GenAI for simulating healthcare scenarios, aiding in decision-making processes and enhancing clinical skills. In this discipline, GenAI technologies are used to create immersive simulations where nursing students can practise responding to diverse patient situations, ultimately improving their ability to make informed decisions in real-world healthcare settings. For example, Liu et al. (2023) conducted a review of ChatGPT's application in nursing education. Their study highlights ChatGPT's potential contributions to nursing education, emphasising simulation scenarios, immediate feedback and the reduction of educator workload. Liu et al.'s (2023) study underlines the potential benefits of leveraging ChatGPT in nursing education while emphasising the importance of addressing challenges and limitations. The research encourages a thoughtful and informed approach for nursing educators considering the integration of ChatGPT into the educational framework for nurses. The application of GenAI in these contexts demonstrates its ability to simulate real-world scenarios, enhance decision-making skills, and bridge the gap between theoretical knowledge and practical application.

Theme 3: A tool for professional skills and decision-making

This theme centres on the integration of GenAI to develop practical skills and facilitate data-driven decision-making processes within professional education disciplines. The theme was dominated by publications classified within 'Business and Management Education' and 'Statistics and Data Science Education'. In the domain of 'Business and Management Education', GenAI is utilised for strategic planning simulations, leadership skill development, market and trend analysis, financial modelling and business scenario analysis. For example, Ratten and Jones (2023) discussed the implications of ChatGPT for management educators, emphasising its ability to impact professional decision-making skills. However, this claim lacks objectivity as the study was conceptual in nature.

Within 'Statistics & Data Science Education', GenAI is employed to teach advanced analytics, automate data processing and enhance statistical modelling skills. Students and educators in this discipline use GenAI-powered tools to explore complex datasets, develop predictive models and gain proficiency in applying statistical methods to real-world scenarios, aligning theoretical knowledge with practical expertise. One prominent study in this discipline was conducted by Xing (2024) who explored the application of ChatGPT in the context of statistics and data analytics. One of the key findings indicates that ChatGPT serves as a valuable aid in addressing conceptual problems and implementing analytical techniques. This functionality enhances the learning experience, making data analysis more accessible and understandable for students.

Theme 4: A writing assistant and productive tool

This theme focuses on the application of GenAI in academic writing and productivity. Disciplines dominating this application include medical education, language education, general education and computing. In computing, Alghamdi (2024) investigates the impact

of ChatGPT-generated feedback on the technical writing skills of computing students. The study reveals emotional, psychological and educational effects, highlighting the potential of ChatGPT in education while emphasising the need for balanced and personalised feedback approaches. In language education, Boudouaia et al. (2024) examines the use of ChatGPT-4 in enhancing EFL writing learning. The study demonstrates significant improvements in students' writing skills and attitudes toward ChatGPT-4, emphasising its potential usefulness in EFL writing practice. Within general education, Dergaa et al. (2023) delves into the prospects and potential threats of ChatGPT in academic writing. While acknowledging its potential to enhance efficiency, concerns regarding the authenticity and credibility of academic work are raised, underscoring the importance of ethical considerations and human intelligence in the research process. In summary, these studies collectively underscore the potential of GenAI in enhancing writing skills and productivity across various disciplines. However, they also highlight the importance of ethical considerations, responsible usage, and the need for human intelligence to complement GenAI's capabilities in academic writing contexts.

Theme 5: An interdisciplinary tool

Although the above four themes were dominated by specific research disciplines, this theme is dominated by a wide range of disciplines, ranging from general education to a more specific discipline. Publications within this theme exhibit several inherent characteristics, including interdisciplinary focus, diversity of application, innovation and integration, ethical considerations and real-world impact. The goal of these studies is to uncover the shared applications and impacts of GenAI that extend beyond specific educational, professional and research domains.

For example, Dwivedi et al.'s (2023) work sheds light on the multidisciplinary perspectives of ChatGPT. The authors bring together insights from 43 experts spanning various fields such as computer science, marketing, information systems, education, policy, hospitality, tourism, management, publishing and nursing. This paper explores the transformative nature of ChatGPT, emphasising its potential across diverse contexts. In the article, GenAI is seen as a double-edged sword, offering opportunities for enhanced productivity, particularly in banking, hospitality, tourism and information technology. However, it also poses ethical and legal challenges and has the potential for negative impacts, including disruptions to practices, threats to privacy and security, and consequences related to biases, misuse and misinformation.

Nikolic et al. (2023) conducted a multidisciplinary and multi-institutional benchmarking and analysis of ChatGPT. The study responds to the significant impact of ChatGPT, which raised questions about assessment integrity and the potential challenges in detecting plagiarism within higher education. The authors delve into the effects of ChatGPT on engineering education assessment by examining its responses to existing assessment prompts from ten subjects across seven Australian universities. The analysis explores both the strengths and weaknesses of current assessment practices while discussing potential opportunities for ChatGPT to facilitate learning and the development of critical thinking skills. The findings highlight the need for changes in current assessment practices, as ChatGPT, with minimal modification to input prompts, could generate passable responses to many assessments.

Within the diverse studies conducted across various thematic areas, it is clear that GenAI technologies have made substantial advancements in different educational domains. These studies reveal a complex finding, highlighting the potential advantages as well as the challenges linked to the incorporation of GenAI in teaching and learning practices. However, our analysis suggests that disciplinary characteristics prompt authors to have a specific focus on their contribution to GenAI technologies.

DISCUSSION

We conducted a comprehensive review of publications discussing the application of GenAI in education. Our review indicated a robust and growing interest in GenAI research between 2023 and 2024. The implications of this sustained interest are far-reaching. It could influence the research agenda for subsequent years (Labadze et al., 2023), guiding future directions and priorities within the GenAI community. Furthermore, the findings from these publications may impact policy and practice and inform decision-making processes for policy makers and practitioners. The adoption of AI technologies in education is dynamic and evolving, and the knowledge generated from these publications could serve as a crucial resource for those shaping educational policies and implementing AI-driven solutions in educational practices.

Our review found that GenAI technologies have emerged as a focal point of extensive publications and discussions spanning a multitude of educational disciplines. Consistent with prior reviews (Ansari et al., 2023; Bahroun et al., 2023; Ipek et al., 2023), this inclusive reach encompasses a wide range of disciplines. The versatility of GenAI's applications has led to a cross-disciplinary engagement, with scholars and practitioners from diverse educational domains actively contributing to the discourse. These contributions take various forms, ranging from original research articles and reviews to commentaries, editorials, letters and opinions (see also Bahroun et al., 2023; Kim et al., 2023; Memarian & Doleck, 2023). The diversity in publication types underscores the multifaceted nature of GenAI's impact on education. Original research articles provide empirical insights and findings, reviews offer comprehensive analyses of existing literature, and commentaries and editorials provide critical perspectives and reflections on the implications of GenAI in educational contexts. Letters and opinions contribute to a dynamic dialogue, offering personal insights and perspectives on the evolving landscape of GenAI in education.

The present study found that, consistent with prior reviews (Bahroun et al., 2023), academic publications on GenAI span diverse discursive themes, reflecting a comprehensive exploration of its societal impact and applications. These research areas include 'application, impact and potential', 'ethical implication and risks' and 'perspectives and experiences'. Additionally, a significant focus has been directed toward 'institutional and individual adoption', 'performance and intelligence', 'ethical guidelines and framework', 'attitude, awareness, and motivation' and 'evolution and trends'.

While higher education and general education monopolise the academic discourse on GenAI's impact (Strzelecki, 2023), the sheer abundance of publications in these domains raises scepticism. This overwhelming trend suggests more of a collective rush than genuine depth, indicating a pervasive interest in exploring GenAI's purported influences on learning. However, this prolific output might also signify conformity rather than critical inquiry, potentially obscuring a thorough examination of the actual implications and challenges associated with the widespread adoption of GenAI technologies in universities and advanced learning institutions.

The glaring omission in the discourse surrounding GenAI becomes striking when examining its negligible presence in the context of K-12 education. The conspicuous lack of publications at these educational levels doesn't just suggest a potential research gap; it underscores a significant oversight, emphasising the urgency for a more profound exploration into the effective integration of GenAI into K-12 education. The failure to understand the challenges and opportunities specific to K-12 education perpetuates educational inequality and also reflects a myopic focus within the academic community. Su and Yang's (2023) attempt to draw attention to this deficiency is commendable, but it also serves as a stark reminder of the prevailing neglect. Similarly, Zhang and Tur (2023) observed this disparity and were forced

to include only 13 articles (related to K-12) in their review while Hays et al. (2024) assert that little research has been explored to understand what K-12 teachers think about GenAI. As the field evolves, rectifying this identified gap is not just an academic exercise; it is imperative for educators, policy makers and researchers aiming to exploit GenAI's potential across the entire educational spectrum. A genuinely comprehensive and critical approach to GenAI research should go beyond lip service, considering its implications for learners at all stages of their academic journey.

We found that GenAI is conceptualised differently across disciplines. Our analysis revealed five themes that explained these conceptualisations, including the application of GenAI as a tool for pedagogical enhancement (Barrot, 2023), where it transforms teaching and learning experiences through personalised approaches, adaptive methodologies and interactive content. Additionally, GenAI was conceptualised as a tool for specialised training and practice (Totlis et al., 2023), addressing the unique needs of disciplines such as medical and nursing education by simulating real-world scenarios and enhancing decision-making skills. Furthermore, GenAI is employed for professional skills and decision-making (Ratten & Jones, 2023), playing a vital role in business and management education for strategic planning simulations and in statistics and data science education for teaching advanced analytics. GenAI is also conceptualised as a writing and productive tool (Boudouaia et al., 2024), addressing the unique needs of disciplines such as language and programming education and general academic writing. Lastly, the interdisciplinary application theme explores GenAI's versatile and interdisciplinary nature, spanning various fields and emphasising both its potential benefits and the ethical considerations associated with its widespread use (Dwivedi et al., 2023). Overall, we propose that different educational disciplines have preferred and unique ways of applying GenAI in education and research.

SUMMARY OF GAPS IN THE LITERATURE

A paucity of research and discussions on GenAI in K-12 education

As our findings suggest, there is a paucity of research discussing GenAI applications in K-12 education. Several factors may contribute to this limited attention. Firstly, there might be a deliberate focus on higher education in the current research context, potentially overlooking the distinctive requirements and dynamics of K-12 education. Secondly, technical challenges in implementing GenAI at the K-12 level, such as issues related to infrastructure, accessibility and age-appropriate content, could be hindering research in this domain (Felix & Webb, 2024). Additionally, there might be a lag in the integration of advanced technologies in K-12, further influencing the level of attention given to GenAI in these educational settings.

The consequential nature of this gap becomes apparent when considering its implications. K-12 education serves as the bedrock for students' academic journeys, and the potential influence of GenAI on teaching methods, learning outcomes and educational experiences during these formative years is substantial. The lack of research in this critical area hinders the development of effective strategies for responsibly integrating GenAI into K-12 education and raises concerns about the haphazard implementation of these technologies. However, it is important to note that some research in K-12 education might be published in grey literature, which could provide additional insights not captured in our review.

A limited exploration of GenAI impact using experimental procedures

In light of the extracted keywords and our comprehensive understanding of research methodologies, it is crucial to assert that the potential and impact of GenAI technologies have not been adequately scrutinised through experimental procedures. Supporting this view, Zawacki-Richter et al. (2024) in a recent editorial concluded that the knowledge of the actual impact of AI technologies is insufficient. We propose that knowledge of the impact of AI technologies on education can be provided through experimental means. Notably, within the corpus of the extracted articles, only a limited number of articles employed an experimental approach (see Sun et al., 2024; Yilmaz & Yilmaz, 2023). This paucity of experimental studies raises critical questions regarding the comprehensive exploration and validation of the impacts of GenAI technologies. It underscores a significant research gap that necessitates further investigation and systematic exploration.

Potential and ethical concerns of GenAI are not discussed from the lens of cultural dimensions

The limited exploration of GenAI from a cultural dimension underscores another crucial gap in our understanding, particularly given the well-established recognition that culture plays a pivotal role in shaping attitudes and behaviours, including the acceptance of technology (Jan et al., 2022; Sun et al., 2019). Existing studies on technology acceptance (see e.g., Ivanov et al., 2024; Niloy et al., 2024; Strzelecki, 2023) consistently neglect cultural influence as opposed to recent and previous findings indicating the influence of cultural factors on how individuals and societies embrace new technological innovations (see Tarhini et al., 2017; Yusuf et al., 2024). Acknowledging this, an exploration of the potential and concerns of GenAI within the context of culture becomes imperative. Understanding the cultural dimension of GenAI is essential as it can provide deeper insights into how GenAI is perceived, adopted and utilised.

A paucity of interdisciplinary perspectives

We identified another notable research gap that warrants attention. While extensive research has been conducted in various educational contexts, our review suggests that there is limited empirical or theoretical evidence examining how GenAI can be integrated across multiple academic disciplines concurrently. This observation is not intended to diminish the existing contributions within specific disciplines but rather to advocate for a broader and more interconnected approach to GenAI research in education. Embracing interdisciplinary perspectives can uncover insights that contribute to a more comprehensive understanding of GenAI's implications. This critical perspective aligns with the growing recognition that complex issues often necessitate interdisciplinary approaches (Newman, 2023). Understanding the implications of GenAI in education requires such an interdisciplinary lens.

CONCLUSION

The emergence of AI technologies has led to extensive research within the educational domain. However, amidst this growing body of research, a crucial aspect has been notably scarce—an assessment of thematic focus and conceptualisation within GenAI research.

This scholarly discourse undertook a thorough examination of publications related to GenAI technologies, highlighting research and discursive patterns to uncover their transformative potential and concerns, while simultaneously exploring the gaps in the literature. GenAI's potential is evident, and it attracts varied research focus and conceptualised applications. The emphasis on higher education suggests a robust interest, while a notable gap in K-12 education underscores the need for further exploration. This transformative technology offers promising benefits, but responsible use and ethical considerations are crucial for navigating associated challenges. A balanced approach is essential to maximise GenAI's positive impact on the educational experience.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The dataset for this study can be accessed by following these links: <https://doi.org/10.17605/OSF.IO/WSDAJ> <https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=605>.

ETHICS STATEMENT

The review adhered to the PRISMA 2020 recommendations in synthesizing and reporting the evidence.

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ENDNOTES

ⁱ The PRISMA checklist for this review can be found in Open Science Framework (OSF) here: <https://doi.org/10.17605/OSF.IO/WSDAJ>.

ⁱⁱ <https://doi.org/10.17605/OSF.IO/WSDAJ>.

ⁱⁱⁱ See the full article database <https://doi.org/10.17605/OSF.IO/WSDAJ>.

^{iv} See article mapping extraction codes <https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=605>.

^v At the time of writing, the current version is 6.15.3.0. See <https://eppi.ioe.ac.uk/EPPIReviewer-Web/home>.

^{vi} <https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=605>.

^{vii} In the EPPI Visualizer, go to the EPPI Mapper section and click on the 'view map' against the 'Research area By Discipline' map. <https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=605>.

^{viii} In the EPPI Visualizer, go to the EPPI Mapper section and click on the 'view map' against the 'Conceptualization By Discipline' map. <https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=605>.

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APPENDIX A

Article distribution by type

Article type	Count	Percent
Original research	323	79.4
Review	33	8.1
Commentary	9	2.2
Editorial	10	2.5
Opinion	20	4.9
Letter	7	1.7
Discussion	5	1.2

APPENDIX B

Article distribution by educational level

Educational level	Count	Percent
K-12	13	3.2
General education	88	21.6
Higher education	306	75.2

APPENDIX C

Article distribution by discipline

Discipline	Count	Percent
Accounting	2	0.5
Business education	1	0.2
Chemistry education	5	1.2
Computing/Engineering	21	5.2
Computing education	10	2.5
Early childhood education	6	1.5
General education	220	54.1
Geography education	1	0.2
Language education	32	7.9
Legal education	2	0.5
Management education	3	0.7
Medical education	54	13.3
Nursing education	13	3.2
Multidisciplinary	3	0.7
Data science	2	0.5
Physical and health education	2	0.5
Physics	2	0.5
Physics education	2	0.5
Primary education	1	0.2
Science education	2	0.5
STEM	3	0.7
Tourism education	3	0.7
Others	17	4.2

APPENDIX D

Article distribution by focus/area

Research focus	Count	Percent
Attitude, awareness and motivation	16	3.9
Ethical guidelines and framework	28	6.9
Ethical implication and risks	85	20.9
Application, impact and potential	242	59.5
Institutional and individual adoption	57	14.0
Performance and intelligence of GenAI	40	9.8
Perspectives and experiences	70	17.2
Evolution and trends	10	2.5

Note: Some articles have more than one research focus/area.

APPENDIX E

Cross-tabulation of discipline and research focus

Discipline	Attitude, awareness and motivation	Ethical guidelines and framework	Ethical implication and risks	Application, impact and potential	Institutional and individual adoption	Performance and intelligence	Perspectives and experiences	Evolution and trends
Accounting	0	0	1	1	0	1	0	0
Biology education	0	0	0	0	0	0	0	0
Business education	0	0	0	1	0	0	0	0
Chemistry education	0	0	0	4	0	2	1	0
Computing/Engineering	0	1	2	17	1	4	4	0
Computing education	0	0	0	6	0	3	4	0
Early childhood education	0	0	1	4	0	0	2	0
Entrepreneurship education	0	0	0	0	0	0	0	0
General education	8	15	41	113	46	14	39	9
Geography education	0	1	0	0	0	0	0	0
Language education	3	4	4	22	1	1	7	0
Legal education	0	0	0	2	0	0	0	0
Management education	0	1	1	2	0	0	1	0
Medical education	5	3	21	31	4	8	8	0
Nursing education	0	1	6	10	1	1	1	1
Nuclear medicine education	0	0	0	0	0	0	0	0
Multidisciplinary	0	0	0	3	0	0	0	0
Measurement and Evaluation	0	0	0	0	0	0	0	0
Data science	0	0	0	2	0	1	0	0
Physical and health education	0	0	2	2	0	0	0	0
Physics	0	0	0	0	0	1	1	0
Physics education	0	0	0	2	1	1	0	0
Primary education	0	0	0	0	0	0	1	0
Science education	0	0	1	2	1	1	0	0
STEM	0	0	0	3	0	0	0	0
Tourism education	0	0	2	2	0	1	0	0
Others	0	2	3	12	2	1	1	0

APPENDIX F

Conceptualisation of GenAI application

Conceptualisation	Count	Percent
A tool for pedagogical enhancement	206	50.6
A tool for specialised training and practice	72	17.7
A writing assistant and productive tool	78	19.2
A tool for professional skills and decision-making	16	3.9
An interdisciplinary tool	210	51.6

Note: Some articles conceptualise GenAI applications in more than one category.

APPENDIX G

Cross-tabulation of discipline and conceptualisation of GenAI application

Discipline	A tool for pedagogical enhancement	A tool for specialised training and practice	A writing assistant and productive tool	A tool for professional skills and decision-making	A cross-cutting technology
Accounting	0	0	0	2	1
Biology education	0	0	0	0	0
Business education	0	0	0	1	0
Chemistry education	4	1	1	1	1
Computing/Engineering	14	2	9	0	8
Computing education	8	0	6	2	2
Early childhood education	3	0	1	1	2
Entrepreneurship education	0	0	0	0	0
General education	131	4	26	3	159
Geography education	0	0	0	0	1
Language education	22	0	21	0	4
Legal education	0	1	0	0	2
Management education	0	0	0	3	1
Medical education	7	50	8	0	10
Nursing education	0	10	1	0	4
Nuclear medicine education	0	0	0	0	0
Multidisciplinary	0	0	0	0	4
Measurement and Evaluation	0	0	0	0	0
Data science	0	0	1	2	0
Physical and Health education	1	1	0	0	1
Physics	2	0	0	0	0
Physics education	1	1	0	0	0
Primary education	1	0	0	0	0
Science education	2	0	0	0	0
STEM	3	0	0	0	2
Tourism education	1	0	1	0	1
Others	6	2	3	1	7